

1 Condensation coding

In this section, we discuss *condensation coding*, the communication-theoretic version of the condensation problem. The condensation coding problem is to communicate $(X_i)_{i \in I}$ in a distributed manner, with a message Y_A for each $A \in \mathcal{P}^+I$; we ask that X_i is recovered from the messages Y_A with $i \in A$ (but the encoder for Y_A can see all X_i , not just those with $i \in A$); furthermore, communication is to be performed jointly for a batch of size $n \in \mathbb{N}$, requiring recovery with probability $1 - o_{n \rightarrow \infty}(1)$. Except for allowing some small probability of error, the condensation coding problem for $(X_i)_{i \in I}$ is just the condensation problem for the variables $X'_i = (X_i^1, \dots, X_i^n)$ for $i \in I$. However, the questions we examine in this section differ from questions about the condensation of random variables examined earlier also in that we will be interested in behavior in the limit $n \rightarrow \infty$ and we will now be interested in how long the messages need to be for communication to be possible — this is like replacing our earlier complexity measures based on Shannon entropies $H(Y_A)$ of the latents with complexity measures based on their Hartley entropies, i.e., $(\log_2 \text{ of})$ the sizes of their domains.

definition. what it is to condensation-code some ground variables at some capacities

Let's say a $(n, \mathbf{r}, \varepsilon)$ condensation-coding of the variables $(X_i)_{i \in I}$

proposition: getting examples for kolmogorov condensation from coding

discussion of how Wyner's common information says something about the two-variable case, and of whether there could be a single-letter characterization of